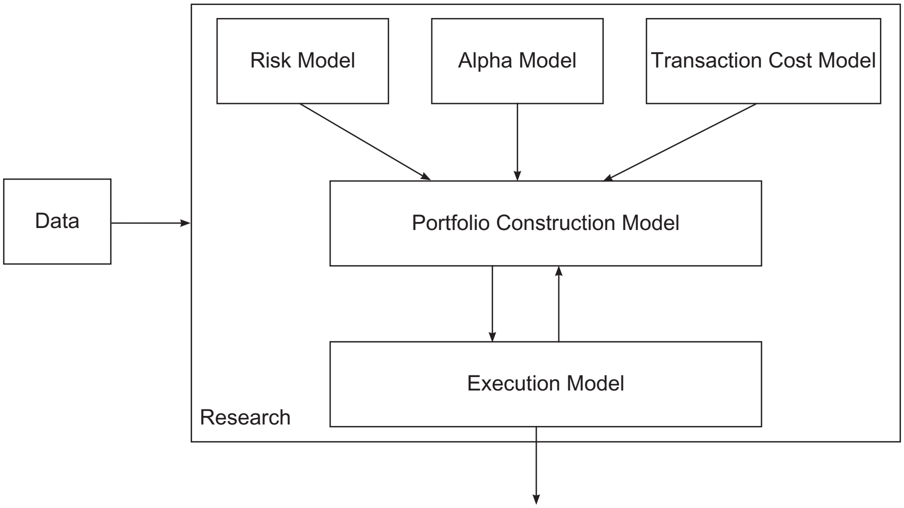
a general stabilization of market systems. Though they are premised in the antithesis of efficient markets, high frequency strategies remove market inefficiencies and impound information into prices faster. Liquidity is enhanced, owing to the increased trade count; however, during severe dislocations this may become scarce. In addition, automation provides numerous operational benefits, i.e., a reduced headcount and thus reduced expenses alongside removal of human error.

Frequently, the term black box trading is used to describe high frequency trading strategies. Black box trading or black box models in general are seemingly complex and mathematically sophisticated ways of detecting anomalies and inefficiencies in the market. Indeed, quantitative trading thrives through such a façade; however; the innards of such models are not beyond the understanding of most people. Quantitative trading applies a rigorous, thorough and scientific method to the trading process. These are the inner workings of the black box, which are commonly understood to be unknown and unknowable. Figure 13.1 removes the illusory veneer from the black box. It is important to note that such a setup is not the same across traders. However, it succinctly captures the key groupings within their frameworks (Narang, 2009).

Inside the black box there are three key components: the alpha model, the risk model, and the transaction cost model, which feed into the portfolio construction model, which in turn feeds into the execution model. The alpha model is the edge sought by quant traders, being the focus of this



■ **Figure 13.1** Stylized Black Box Trading System